

$$7) \lim_{x \rightarrow 1} -\frac{x^2 - 1}{x - 1}$$

$$8) \lim_{x \rightarrow 5} -\frac{x^2 - 5x}{x - 5}$$

$$9) \lim_{x \rightarrow 2} -\frac{x^2 - x - 2}{x - 2}$$

$$10) \lim_{x \rightarrow -5} \frac{x^2 + 3x - 10}{x + 5}$$

$$11) \lim_{x \rightarrow 0} \frac{\frac{1}{-4 + x} + \frac{1}{4}}{x}$$

$$12) \lim_{x \rightarrow -3} \frac{x}{\frac{1}{3 + x} - \frac{1}{3}}$$

$$13) \lim_{x \rightarrow 5} \frac{x - 5}{\sqrt{x + 4} - 3}$$

$$14) \lim_{x \rightarrow 3} \frac{\sqrt{x + 6} - 3}{x - 3}$$

Critical thinking questions:

15) Give an example of a limit of a rational function where the limit at -1 exists, but the rational function is undefined at -1.

16) Give two values of a where the limit cannot be solved using direct evaluation. Give one value of a where the limit can be solved using direct evaluation.

$$\lim_{x \rightarrow a} \frac{x}{\frac{1}{-2 + x} + \frac{1}{2}}$$